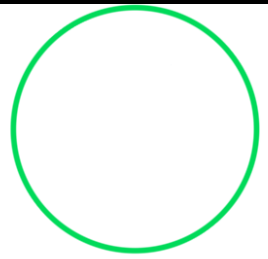


ARRAY : Video -



CodeStorywithMIK



codestorywithmik



CSwithMIK



codestorywithMIK

< WeekendWithMIK



WeekendWithMIK

@WeekendWithMIK

Welcome to WeekendWithMIK



11iK.

They this channel to

see "Life behind the **Scenes + Tech News**

Motivation -

Learning a new skill is like building a bridge while you are still crossing it.

It might feel shaky at first, but every hour of study makes the path stronger.

Keep going until the day you can walk across it with ease.



MIK...

3548. Equal Sum Grid Partition II

Hard

Topics

Companies

Hint

You are given an $m \times n$ matrix `grid` of positive integers. Your task is to determine if it is possible to make **either one horizontal or one vertical cut** on the grid such that:

- Each of the two resulting sections formed by the cut is **non-empty**.
- The sum of elements in both sections is **equal**, or can be made equal by discounting **at most** one single cell in total (from either section).
- If a cell is discounted, the rest of the section must **remain connected**. ←

Return `true` if such a partition exists; otherwise, return `false`.

Note: A section is **connected** if every cell in it can be reached from any other cell by moving up, down, left, or right through other cells in the section.

Example :

1	2	4
---	---	---

2	3	5

Output: False

Sec-1	Sec-2
1	2
3	4
4	$6 - 2 = 4$

Output: True

Constraints:

- $1 \leq m == \text{grid.length} \leq 10^5$
- $1 \leq n == \text{grid}[i].\text{length} \leq 10^5$
- $2 \leq m * n \leq 10^5$
- $1 \leq \text{grid}[i][j] \leq 10^5$

Thought Process

	0	1	2	3	
0					top
1					10
2					6

total

top

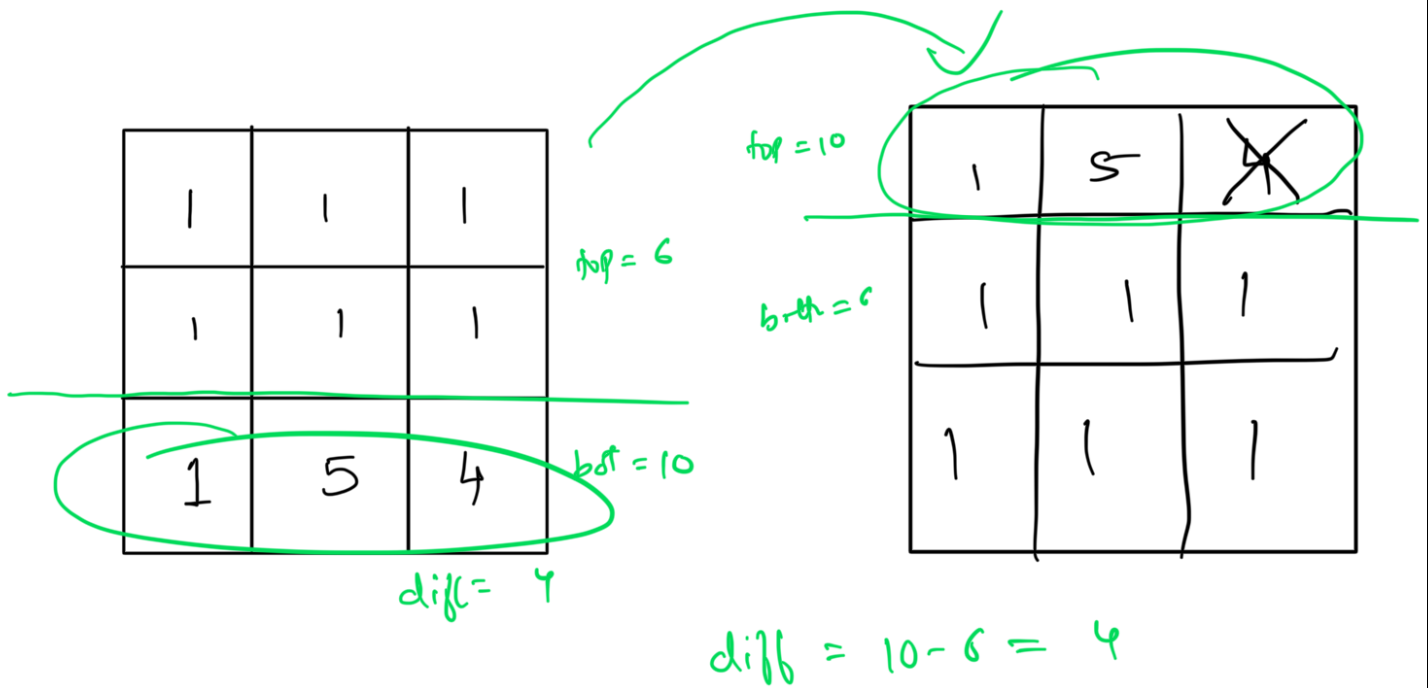
bottom = total - top

3				
4				

$$\text{diff} = 10 - 6 = 4$$

$\text{top} == \text{bottom} \rightarrow \text{True.}$

$$\text{diff} = \boxed{\text{top} - \text{bottom}} = 10 - 6 = 4$$



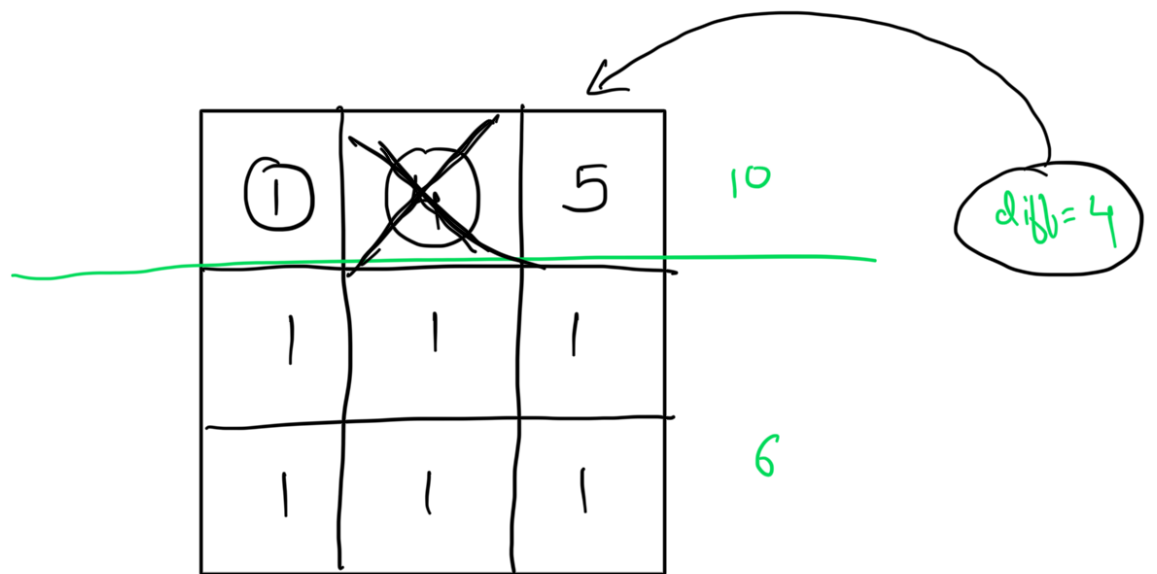
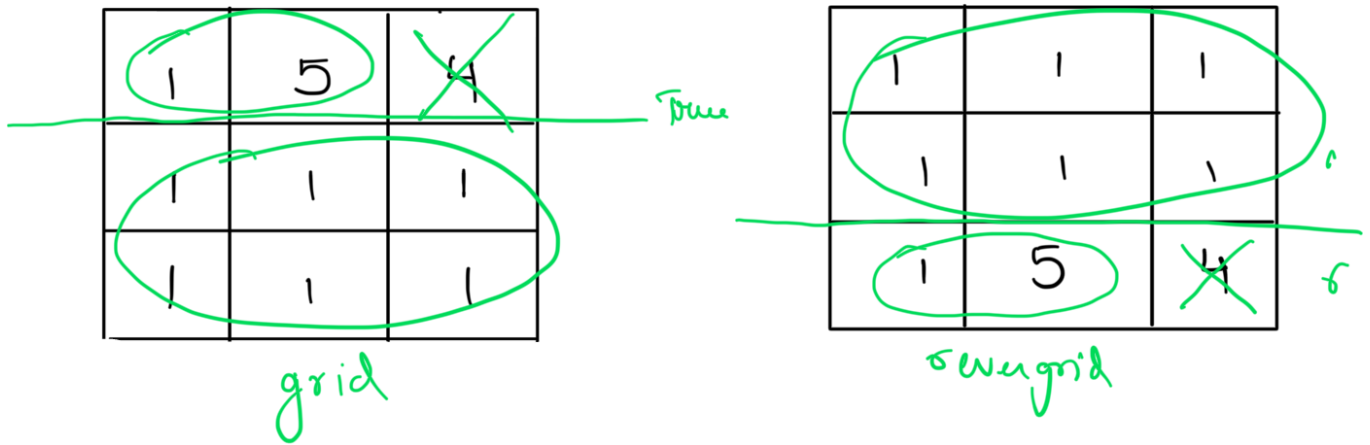
Horizontal Cut

- grid
- ① Traversal top to bottom ✓✓
 - ② Set <int> $\rightarrow$ top ⁹ elements
 - ③ top
 - ④ bottom = total - top;
 - ⑤ diff = $\boxed{\text{top} - \text{bottom}}$

Search in the top section.

remove if found
& check connectivity

reversing the grid.



How to check connectivity
of cells after removing

a cell ???

	0	1	2	3	4	
0	a	<u>b</u>	<u>c</u>	<u>d</u>	e	top
i = 1	<u>f</u>	<u>g</u>	<u>h</u>	<u>i</u>	<u>j</u>	
i = 2						
3						bottom
4						

m x n m = 5, n = 5

$$\text{diff} = \text{top} - \text{bottom}$$

If top section $\rightarrow$ 1 row $\rightarrow$ deleting only
 $[0][0]$ or $[0][n-1]$

$$\text{Safe cells} = \underbrace{[0][0]}_{\text{diff}}, \underbrace{[0][n-1]}_{\text{diff}}$$

if (diff == 0) $\rightarrow$ True

if (diff == grid[0][0]) $\rightarrow$ True

if (diff == grid[0][n-1]) $\rightarrow$ True

if (diff == grid[i][0]) $\rightarrow$ True

if ($i > 0 \ \&\& \ n > 1 \ \&\& \ \text{Set.Count}(\text{diff})$) $\rightarrow$ True

0	10
1	5
2	4
3	5

0	5	
1	5	
2	5	top = 14
3	10	bottom = 10

diff = 4

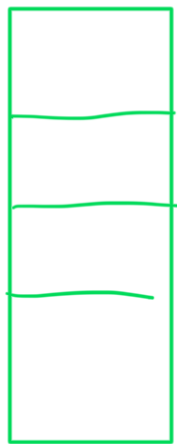
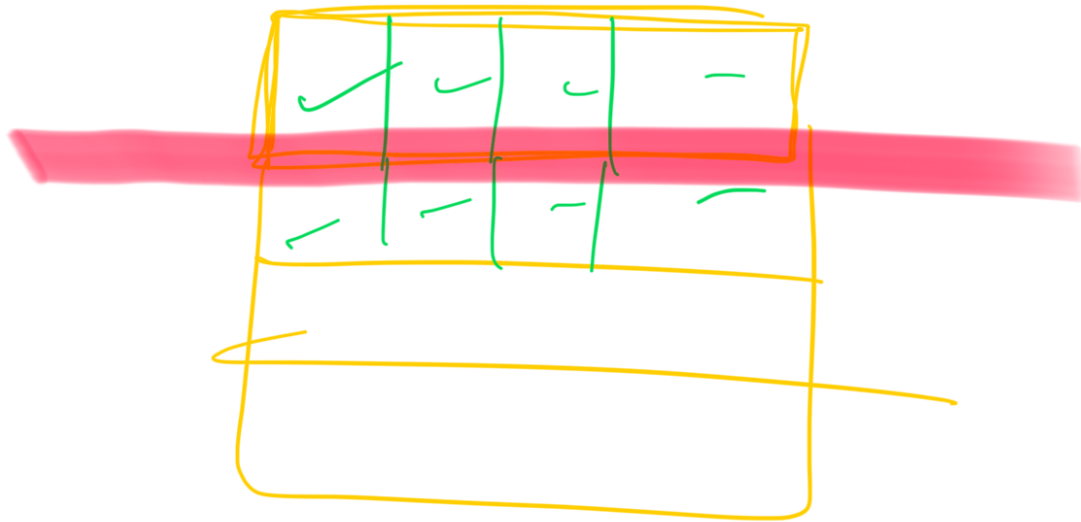
$m = 4$
 $n = 1$

Set = count (4)
Set = { 5, 4 }

	0	10	
0		10	
1			
2		10	

top = 14
bottom = 10
diff = 4
set = { 10, 4 }

[i][0]



Horizontal Cut

```
boolean checkHorCut( grid) {
```



```

m = grid.size();
n = grid[0].size();

```

```

unordered_set<int> st;
top = 0;
for (i = 0; i < m - 2; i++) {

```

```

    for (j = 0; j < n; j++) {
        st.insert(grid[i][j]);
        top += grid[i][j];
    }

```

```

    bottom = total - top;
    diff = top - bottom;
    if (diff == 0) True;

```



```

    if (diff == grid[0][0]) True;
    if (diff == grid[0][n-1]) True;
    if (diff == grid[i][0]) True;
    if (i > 0 && n > 1 && seen.count(diff))
        True;

```

```

    return false;
}

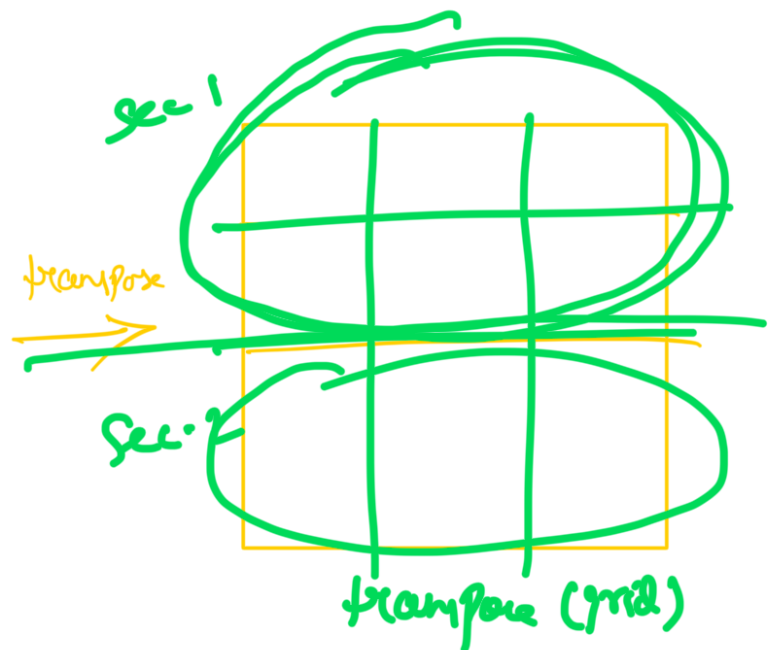
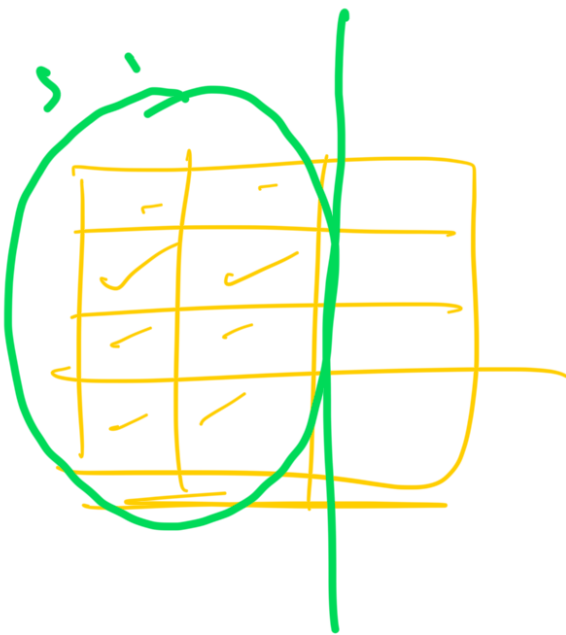
```

4

```
if (checkHorCut (grid))  
    return True;
```

```
reverse (begin(grid), end(grid));
```

```
if (checkHorCut (grid))  
    return True;
```



// transpose grid

```
if (checkHorCut (grid))
```

19000

✓ reverse

✓ if (arr. (grid))